Attention-Based Sequence Models: Notes

Abstract. Recurrent networks read a sentence one word at a time, which makes long-range dependencies hard to learn and training hard to parallelise. Attention lets every position look at every other position directly. A model built only from attention layers and feed-forward layers trains faster and translates better than recurrent models of the same size.

1. Scaled dot-product attention
Each position produces a query, a key and a value. The output for a position is a weighted average of all values, with weights softmax(Q K^T / sqrt(d)). Dividing by the square root of the key dimension keeps the softmax from saturating.

2. Multi-head attention
Several attention heads run in parallel on smaller projections, so each head can attend to a different kind of relation.

3. Positional information
Attention ignores word order, so the model adds positional encodings to the inputs: sine and cosine waves of different frequencies.

4. Cost
Attention compares every pair of positions, so its cost grows with the square of the sequence length.